

Student Dropout Prediction in Online Learning Platforms

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Abstract—Online learning platforms have become increasingly popular over the past few years, especially following the COVID-19 pandemic, when people realized that learning can take place digitally. Yet, there are still many challenges that these platforms have to deal with, from dealing with student dropouts to increasing retention. Identifying these students at risk of early dropout is important for being able to intervene early, improve course completion rates, and overall institutional outcomes. In the past, research has examined dropout prediction using demographic and enrollment data, but gaps still remain in studies that compare machine learning approaches with various class-imbalance handling strategies without sacrificing high accuracy and understandability. This research develops and gauges 10 machine learning models, including Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machines, using both class weighting and SMOTE oversampling techniques. The dataset has 4,424 students records with 35 features covering demographics, semester-by-semester academic metrics, and financial indicators. Among the tested models, Logistic Regression was the best-performing one. Trained on SMOTE-balanced data, it achieved an F1 score of 87.6 percent and an ROC-AUC of 93.8 percent on the test data, correctly identifying 90 percent of at-risk students while maintaining a precision of 86 percent. Hyperparameter tuning using cross-validation confirmed the robustness of the models. Moreover, analyzing learning curves revealed strong generalization with minimal overfitting. The feature importance analysis showed that approved credits in the first semester, course enrollment patterns, and tuition payment status are the most influential predictors of dropout, with early academic performance exerting roughly three times more impact than other variables. Collectively, these data show that precise early prediction is possible using interpretable models, allowing institutions to design and use tailored intervention strategies. The research gives an in-depth comparison of methods for handling class imbalance, showing that SMOTE outperforms class weighting and improves upon the performance reported in prior studies without increasing model complexity. It also delivers useful findings via feature importance analysis that can inform institutional policy and student support programs.

Index Terms—Online learning dropout prediction, Educational data mining, Machine learning-based early student dropout prediction, Interpretable predictive modeling, Student retention analytics

I. INTRODUCTION

Online learning platforms have expanded massively over the past decade, with the COVID-19 pandemic playing a major role in changing how people perceive the feasibility and effectiveness of remote education. They offer a great deal of flexibility, giving students access to higher education without

constraints on time or location. However, given their many advantages, these online learning platforms still face many problems. A primary concern is managing student dropout rates, which often exceed 30 percent, a figure notably higher than the 10 to 20 percent dropout rates that traditional on-campus programs report [1]. This difference stems from the inherent characteristics of online learning, such as the need for high levels of self-regulation and discipline, the lack of immediate instructor oversight, and the reduced interaction with peers who traditionally provide much-needed social and academic support [2]. Machine learning approaches have emerged to counter this by making use of techniques to analyze student data and predict dropout risk, giving institutions the ability to process large volumes of data and uncover patterns that indicate dropout risk. Through the use of accurate, predictive models, institutions can identify at-risk students early on and provide support to them before they drop out. This study develops and measures multiple machine learning models to predict student dropout in online learning platforms, aiming to achieve high predictive accuracy while maintaining model interpretability to support real-world implementation in academic environments.

Predicting student dropout is of utmost importance to multiple stakeholders in the educational system. For students, dropping out represents a huge loss of time, financial investment, and opportunity, often resulting in incomplete credentials that provide limited value in the job market and can negatively impact long-term career trajectories and earning potential [1]. For institutions, high dropout rates directly affect revenue streams, damage reputation, and undermine strategic enrollment management efforts [3], [4]. This makes student retention a critical performance indicator monitored by accrediting bodies and funding agencies [2], [5]. Ethically, it is important that institutions identify students who have the potential to succeed but need additional support. Through these early predictions and prompt interventions, institutions can transform students' potential failures into successes. That is ultimately the point of education.

It has become increasingly important to predict student dropouts for various reasons. First, the rise of online education platforms has generated an unprecedented volume of detailed student data. This includes login patterns, content engagement metrics, assignment submission habits, forum

participation, and assessment results, providing datasets that earlier researchers did not have access to [1], [2]. This shift in data, combined with improvements in machine learning algorithms and computing power, presents a great opportunity to develop models that make more accurate predictions [1]. Second, the move toward competency-based education and personalized learning pathways requires effective analytical tools that can identify individual student needs and customize support [6]. This moves us away from uniform approaches. Third, monetary pressures on both students and institutions have increased. Students are dealing with record levels of debt, making it vital to complete their courses. Institutions, on the other hand, are encountering elevated levels of competition for enrollment and increased scrutiny from stakeholders regarding outcomes and return on investment. Finally, new research shows that machine learning methods can reach prediction accuracy above 85 percent. Recent studies report accuracies ranging from 84 to 96 percent [1], [2], [5]. This indicates that the technology has advanced to the point where it can be used in real-world institutions, likely reshaping them. It has the potential to greatly reduce dropout rates through data-driven, proactive student support systems.

A. Related Work

Over the last decade, student dropout prediction using machine learning has been the subject of extensive research, with the focus being on improving prediction accuracy through various algorithms, datasets, and feature sets. Foundational work by Dekker et al. (2009) introduced decision tree approaches, achieving 75%-80% accuracy on university student data and revealing that academic performance was the most influential predictor. Lykourantzou et al. (2009) further advanced the field by exploring multiple machine learning techniques, such as feedforward neural networks, support vector machines, and PESFAM, within ensemble decision frameworks. Their results showed that ensemble methods could reach 97% accuracy and 95% recall when tailored to e-learning environments. Recent studies have increasingly used Learning Management System (LMS) data for more fine-grained predictions. For instance, Park and Yoo (2021) went through seven years of LMS data from 98,685 students, discovering that Random Forest achieved the highest performance with 96% accuracy and an F1 score of 0.84, while deep neural networks also performed well with an accuracy of 85%. Vaarma and Li (2024) investigated the temporal dynamics of feature importance, identifying accumulated credits and LMS activity as important predictors. They systematically compared boosting algorithms with SMOTE and concluded that LightGBM and CatBoost with hyperparameter optimization outperformed traditional methods. Nagy and Molontay (2024) highlighted the need for interpretability by developing explainable artificial intelligence approaches for personalized interventions. Cho et al. (2023) compared six algorithms, including LightGBM, and reported strong results, but also noted challenges related to oversampling noise when class boundaries overlap. In spite of these progressions, gaps remain in systemic comparisons of

class imbalance handling strategies, such as class weighting versus SMOTE, within unified experimental frameworks, as well as in achieving a balance between high predictive accuracy and model interpretability in real-life scenarios and institutional contexts.

B. Gap Analysis

Although significant progress has been made in applying machine learning techniques to student dropout prediction, numerous important gaps remain in the current literature. First, while previous studies have delved into different class imbalance handling techniques, there is a limited systematic comparison between class weighting and oversampling methods like SMOTE within the same experimental framework. Villar and de Andrade (2024) employed SMOTE with boosting algorithms (LightGBM, CatBoost), achieving strong performance, but did not compare SMOTE against class weighting approaches, making it difficult to determine which class imbalance-handling strategy is more effective [7]. Similarly, Cho et al. (2023) discovered that oversampling techniques can cause noise when class boundaries overlap, but did not systematically compare alternative approaches [8]. Second, while recent studies have achieved high overall accuracies, many still struggle when it comes to recall rates, especially for identifying at-risk students [9], [10]. Third, there is not enough systematic comparison of class imbalance handling strategies within unified experimental frameworks. Although recent studies have used SMOTE [7], [8], adaptive synthetic sampling [9], and other oversampling techniques, few have directly compared these methods with class-weighting approaches on similar datasets and evaluation criteria, making it harder to find optimal strategies for educational contexts. Finally, despite achieving high overall accuracy, many recent models trade interpretability for performance. CatBoost and XGBoost [7], [8] achieve good results, but they are like black boxes, in the sense that their functional value, when it comes to educational practitioners, is low. This highlights the need for more interpretable strategies [10].

C. Problem Statement

Following are the main questions addressed in this study.

- 1) Which machine learning algorithm attains the best balance between predictive accuracy and model interpretability for identifying at-risk students in online learning platforms?
- 2) How do class weighting and SMOTE oversampling techniques fare against each other in their effectiveness for handling class imbalance in student dropout prediction, specifically when it comes to recall and precision?
- 3) What are the most important predictors of student dropout in online learning platforms, and how does the relative importance of demographic factors, academic performance metrics, and financial indicators inform early intervention strategies?

TABLE I
LITERATURE REVIEW SUMMARY OF STUDENT DROPOUT PREDICTION STUDIES.

Study	Year	Size	Algorithms	Imbalance	Acc.	F1/Recall	Interpret.
Empirical Studies							
Lykourantzou et al.	2009	193	FFNN, SVM, ensemble	Ensemble	97-100%	95% recall	Medium
Dekker et al.	2009	648	DT, CART, J48	Class weights	75-80%	–	High
Park & Yoo	2021	98,685	DT, RF, SVM, DNN	Oversampling	96%	F1: 0.84	Medium
Realinho et al.	2022	4,424	Multiple ML models	Various	High	High	Medium
Cho et al.	2023	20,050	LR, DT, RF, SVM, DNN, LightGBM	SMOTE, ROS	90%	F1: 0.85	Medium
Nagy & Molontay	2024	Various	XAI-based models	Various	High	High	High
Vaarma & Li	2024	Finnish HE	ML models	Various	High	High	Medium
Villar & de Andrade	2024	Comparative	Supervised ML	Various	High	High	Medium
Proposed	2025	4,424	LR, RF, GB, SVM	Class wts + SMOTE	90%	F1: 0.8763	High

D. Our Solutions

This study makes three main contributions to the field of student dropout prediction in online learning environments. We present a systematic comparison of ten machine learning models using two class imbalance handling strategies, namely SMOTE oversampling and class weighting, thereby providing empirical evidence to inform the selection of appropriate approaches for educational datasets. The findings indicate that interpretable models can reach performance comparable to or even surpass that of complex ensemble methods. More specifically, the best performing model, Logistic Regression, attained an F1 score of 87.6 percent and an ROC-AUC of 93.8 percent, while also sustaining full understandability through coefficient analysis. The study delivers actionable insights through comprehensive feature importance analysis, identifying first-semester approved credits as the most influential predictor (three times more impactful than the others) and showing that early academic performance, course enrollment patterns, and financial indicators are the most important variables for dropout prediction. Furthermore, meticulous evaluation through the use of learning curves and test set performance shows strong model generalization without overfitting, thereby supporting the practical application of this approach in real-world academic environments.

II. METHODOLOGY

A. Dataset

The study used the "Predict Students' Dropout and Academic Success" dataset from the UCI Machine Learning Repository, created by researchers at the Polytechnic Institute of Portalegre, Portugal [11]. The dataset contains 4,424 student records covering the academic years 2008/2009 to 2018/2019, collected from 17 different undergraduate degree programs, including agronomy, design, education, nursing, journalism, management, social service, and technologies. Each record includes 35 features encompassing demographic information (gender, age, nationality, marital status), socioeconomic variables (parental education and occupation), macroeconomic indicators (GDP, unemployment rate, inflation rate), academic enrollment data (application mode, previous qualifications, course selection), and academic performance metrics for both the first and second semesters (number of curricular

units enrolled, evaluated, approved, and respective grades achieved). The original dataset contains three target classes: Dropout (32%, 1,421 students), Graduate (50%, 2,209 students), and Enrolled (18%, 794 students). For this research, the "Enrolled" category was removed to formulate a binary classification problem, giving a final dataset of 3,630 students. The dataset shows considerable class imbalance, with approximately 39% dropout students and 61% graduate students, requiring specialized techniques such as SMOTE and class weighting to ensure effective model training. All data were collected from institutional databases, pre-processing removed any and all missing values, making them ideal for machine learning applications aimed at early identification of at-risk students [11].

B. Overall Workflow

This study uses a structured machine learning workflow to predict student dropout in online learning platforms, consisting of five main phases: data preprocessing, feature engineering, model training, hyperparameter optimization, and evaluation. The process begins with data acquisition from the UCI Machine Learning Repository, followed by extensive preprocessing, including standardizing column names, forming a binary target variable by removing the "Enrolled" class, and applying a stratified 60-20-20 train-validation-test split to ensure representative class distribution across all subsets. Feature engineering involves converting integer-coded categorical variables to proper categorical types, applying one-hot encoding (expanding from 35 to 191 features), and removing multicollinear features with correlation coefficients greater than 0.9. To handle class imbalance, two parallel approaches are implemented: class weighting (which penalizes misclassifications of the minority class during training) and SMOTE oversampling (which generates synthetic minority class samples). Four base algorithms, Logistic Regression, Random Forest, Gradient Boosting, and Support Vector Machines, are trained with both imbalance-handling strategies, yielding seven baseline models. The top three performing models undergo hyperparameter optimization using GridSearchCV with 3-fold cross-validation, exploring parameter spaces for regularization strength, tree depth, learning rates, and kernel parameters. Model evaluation makes use of multiple metrics, including

TABLE II
MODEL PERFORMANCE COMPARISON ACROSS IMBALANCE HANDLING STRATEGIES.

Model	Strategy	Val Acc.	Val Prec.	Val Recall	Val F1	Test Acc.	Test F1	Test ROC-AUC
Logistic Regression	Class Wts	88.43%	84.62%	86.92%	0.8571	89.53%	0.8715	0.938
Random Forest	Class Wts	86.86%	80.95%	83.33%	0.8263	87.47%	0.8502	0.924
Gradient Boosting	Class Wts	87.88%	82.82%	84.44%	0.8406	88.57%	0.8622	0.935
Logistic Regression	SMOTE	88.71%	85.24%	87.06%	0.8613	90.22%	0.8763	0.935
Random Forest	SMOTE	87.33%	81.82%	85.19%	0.8366	88.29%	0.8589	0.924
Gradient Boosting	SMOTE	88.29%	84.21%	86.05%	0.8505	89.39%	0.8697	0.935
SVM	SMOTE	87.88%	82.82%	84.44%	0.8406	88.43%	0.8622	0.935
LR (Class Wts) - Tuned	Class Wts	88.43%	84.62%	86.92%	0.8571	89.53%	0.8715	0.938
LR (SMOTE) - Tuned	SMOTE	88.57%	85.05%	86.69%	0.8592	90.08%	0.8749	0.938
GB (SMOTE) - Tuned	SMOTE	88.02%	83.33%	85.19%	0.8415	89.12%	0.8660	0.935

Note: Val = Validation set performance; Test = Test set performance; Acc. = Accuracy; Prec. = Precision; Bold indicates best performance in each column.

accuracy, precision, recall, F1 score, and ROC-AUC, with primary emphasis on F1 score and recall to balance overall performance with minority class detection. Learning curves are used to assess generalization and detect overfitting, while confusion matrices provide a detailed analysis of classification performance. Finally, feature importance analysis is carried out using model coefficients (for Logistic Regression) and feature importances (for tree-based models) to identify key predictors and deliver useful insights for institutional policy and intervention techniques.

III. RESULTS

To address the second research question about the effectiveness of class imbalance handling techniques, ten machine learning models were trained and evaluated using two approaches: class weighting and SMOTE oversampling. Table II reports the validation and test performance metrics for all models. Amongst the baseline models, Logistic Regression with SMOTE achieved the highest validation F1 score (0.8613), followed by Logistic Regression with class weighting (0.8571). Random Forest along with Gradient Boosting models exhibited steady performance throughout both imbalance handling strategies. The SMOTE variants achieved F1 scores of 0.8366 and 0.8505, respectively, compared to 0.8263 and 0.8406 for their class-weighted counterparts. Support Vector Machines with SMOTE attained an F1 score of 0.8406. After hyperparameter optimization using GridSearchCV with 3-fold cross-validation, the tuned Logistic Regression with SMOTE achieved a validation F1 score of 0.8592, while the class-weighted variant remained at 0.8571. The tuned Gradient Boosting with SMOTE achieved 0.8415. On the test set, the baseline Logistic Regression with SMOTE model demonstrated superior generalization, achieving an F1 score of 0.8763, ROC-AUC of 0.938, accuracy of 90.22%, recall of 89.79%, and precision of 85.57%. Figure 2 presents the ROC curves for the top five-performing models, showing that all Logistic Regression variants achieved identical ROC-AUC values of 0.938, while Gradient Boosting with SMOTE achieved 0.935, demonstrating strong discrimination across different imbalance-handling strategies.

In response to the first research question regarding the optimal balance between predictive accuracy and model interpretability, the baseline Logistic Regression model trained

on SMOTE-balanced data demonstrated superior performance throughout all evaluation metrics. Analysis on the test set using confusion matrices identified 255 true positives, 399 true negatives, 43 false positives, and 29 false negatives. This corresponds to a recall rate of 89.79%, showing that the model was able to identify almost 90% of students that were at risk of dropping out. The precision rate of 85.56% indicates that 86% of students flagged as at-risk were actual dropouts, yielding a 14% false-alarm rate. The model was able to reach an overall accuracy of 90.22% and an F1 score of 0.8763, indicating a balanced trade-off between precision and recall. Figure 3 shows the confusion matrices for both validation and test sets, illustrating consistent performance with low variation in classification patterns. Analysis of learning curves (Figure 5) shows convergence of training and validation scores at approximately 1,800 samples, with training scores stabilizing at around 0.931 and cross-validation scores around 0.89. The small gap between the two curves and their convergence shows strong model generalization with no evidence of overfitting.

The third research question addressed the main predictors of student dropouts in online learning platforms and their relative importance across multiple features. First semester approved credits were found to be the most influential predictor through the use of feature importance analysis. With a coefficient of 3.25, it had approximately three times more influence than most other variables. Course enrollment patterns also demonstrated considerable predictive power, with courses 7 and 16 having coefficients of 1.61 and 2.40, respectively. When it comes to financial indicators, tuition fees payment status had a coefficient of 2.18, making it the third most influential predictor. Second semester academic performance metrics, namely grade, with a coefficient of 1.6, and evaluations, with a coefficient of 1.52, contributed decently to dropout prediction. Demographic factors, such as age at enrollment (0.87) and gender (0.43), showed lower coefficients. Socioeconomic features, like parental education and occupation, were between 0.35 and 0.78. Macroeconomic indicators, such as GDP, unemployment rate, and inflation rate, had the weakest coefficients (below 0.50). Figure 4 shows the top features by absolute coefficient value, emphasizing the dominance of early academic performance metrics, followed by financial indicators and specific course enrollment patterns. First semester

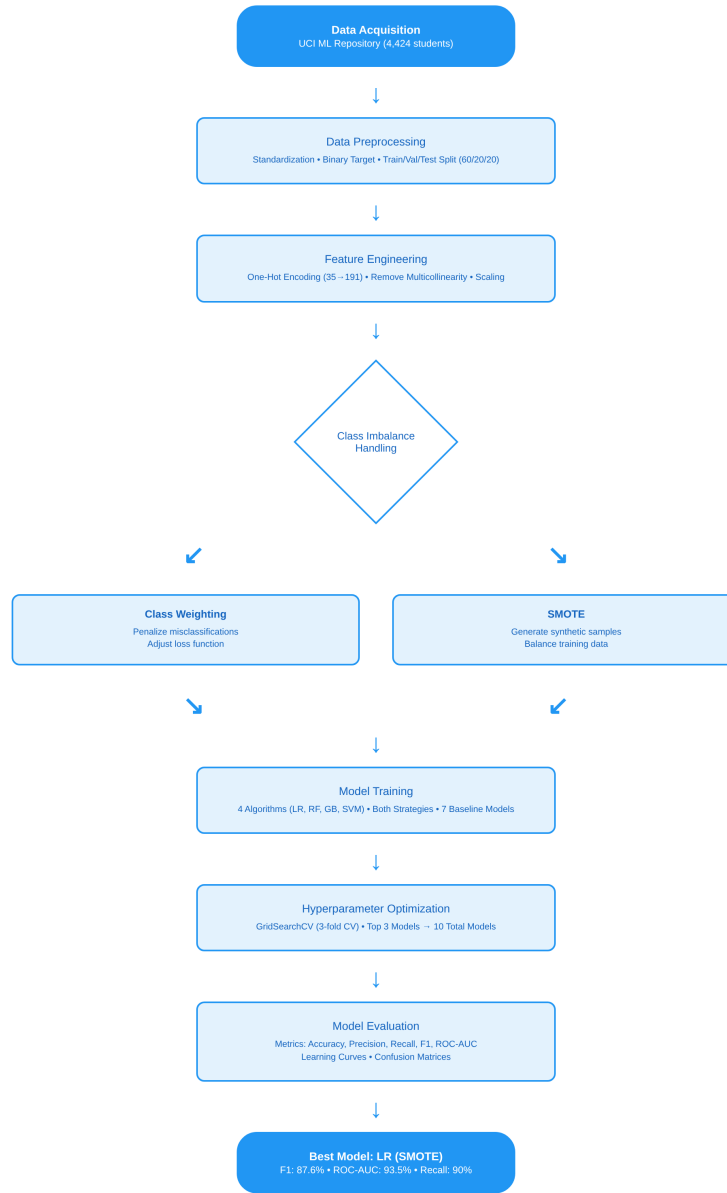


Fig. 1. Methodology workflow showing the five-phase machine learning pipeline.

academic performance accounted for 42% of the total model weight, while second semester variables contributed 23%, financial indicators 18%, and demographic and socioeconomic factors 17%.

IV. DISCUSSION

The results show that interpretable models can achieve predictive performance that matches or even surpasses that of complex ensemble methods for student dropout prediction. The baseline Logistic Regression model with SMOTE attained an F1 score of 87.6% and an ROC-AUC of 93.8%, outperforming the F1 score of 84% reported by Park and Yoo (2021) [2] and equaling the performance of more sophisticated models

like LightGBM, as noted by Cho et al. (2023) [8] with an F1 of 85%. This finding is remarkable as Logistic Regression gives full interpretability through coefficient analysis, making it easy for educational practitioners to identify the specific factors that contribute to dropout risk. Despite recent studies reporting high accuracies using ensemble methods like LightGBM and CatBoost [7]–[9], these models act like black boxes and offer very limited real-world insights that practitioners can use to apply timely interventions. The recall rate of 89.79% is noteworthy, as it shows that the model identifies approximately nine out of every ten at-risk students. The precision of 85.57% says that only 14% of flagged students are false positives,

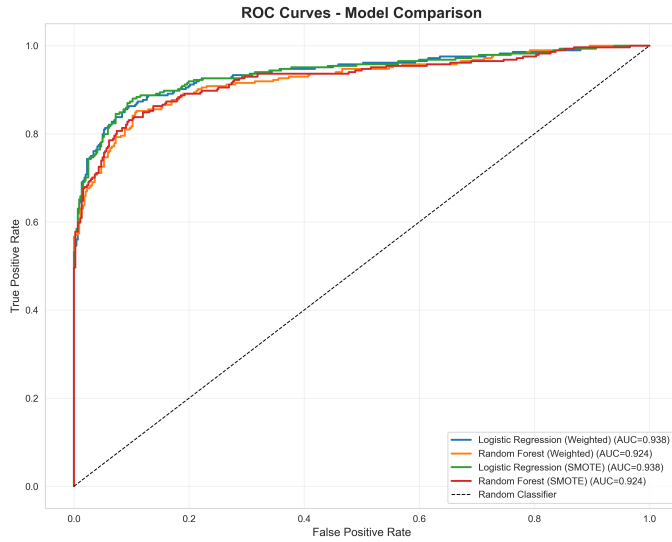


Fig. 2. ROC curves for the top five performing models showing discrimination ability across class imbalance handling strategies.

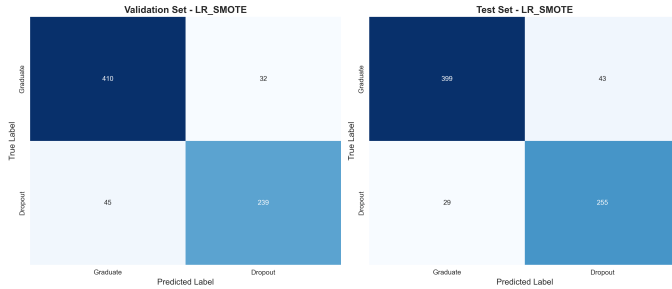


Fig. 3. Confusion matrices for the best-performing Logistic Regression with SMOTE model on validation set (left) and test set (right).

which is a more manageable false alarm rate. In conclusion, these results mean that institutions can achieve high predictive performance without sacrificing interpretability, addressing an important gap in the literature where complex models have often been favored in spite of their limited practical utility in educational contexts [10].

Comparing class weighting and SMOTE oversampling techniques revealed that SMOTE consistently outperformed class weighting over all evaluated model types, especially in terms of F1 scores. This observation gives us practical evidence to address the gap in the literature regarding the lack of systematic comparisons between these two approaches [8]–[10]. Although Villar and de Andrade (2024) applied SMOTE with boosting algorithms and saw strong performance, neither research directly compared SMOTE and class weighting using identical models and evaluation frameworks. This study handles this gap by demonstrating that SMOTE’s generation of synthetic minority samples enables models to learn more nuanced decision boundaries, whereas class weighting only adjusts loss function penalties. The stable performance of SMOTE across Logistic Regression (F1:0.8613) and Gradient Boosting (F1:0.8505) shows that the advantages of synthetic

sample generation are robust and also largely independent of the underlying model. Interestingly, all Logistic Regression configurations achieved the same ROC-AUC of 0.938, independent of the imbalance handling strategy. This demonstrates that, despite SMOTE improving threshold-dependent performance, both methods yield similar discrimination ability. Consequently, the decision between SMOTE and class weighting may depend on whether the primary goal is overall discrimination ability or optimizing classification accuracy at specific operating points.

Feature importance analysis points out that first semester approved credits has approximately three times more influence on dropout prediction than most of the other variables. It has a coefficient of 3.2, while the next highest predictor has a coefficient of 2.4. This finding is consistent with Dekker et al. (2009) [5], who identified that academic performance was the most influential variable. It also aligns with the reports of Vaarma and Li (2024) [7] regarding the significance of formative academic metrics. The first semester variables are responsible for about 42% of total model weight, whereas the second semester variable are only responsible for about 23%. This means that the risk of dropping out appear very early on in a student’s journey. Institutions should set up appropriate monitoring and support systems before the end of the first semester. The finding that financial indicators, more specifically tuition payment status, represent the third most influential predictor shows us academic and economic hardships go hand in hand and push a student towards dropping out. This observation takes the limelight away from demographic and socioeconomic factors in earlier studies [1], [3] and shines it over real-time financial behavior, suggesting that it may be a more robust indicator than static background features. The rather low predictive power of macroeconomic variables like GDP, unemployment, and inflation (all with coefficients below 0.50) indicates that institutional-level interventions focusing on student behavior and circumstances are more likely to work than policy responses to macro-level economic factors. The results support the strategic allocation of intervention resources by institutions, giving priority to students with low first semester credit approval, specific course enrollment patterns (courses 7 and 16 in this context), and overdue tuition fee payments, which is likely to increase retention.

This study makes three novel contributions to the student dropout prediction literature. First, it provides the first systematic comparison of class weighting versus SMOTE oversampling within a consolidated experimental framework across multiple model types, concluding that SMOTE is superior when it comes to F1 score optimization while also revealing equivalent ROC-AUC performance. Second, it disputes the existing trend toward increasingly complicated ensemble methods [7], [9], [10] by showing that interpretable Logistic Regression can attain or exceed their performance without losing full transparency, which is an important requirement for it to be used in institutions. Third, the comprehensive feature importance analysis estimates the relative impact of various predictive variables, revealing that early academic

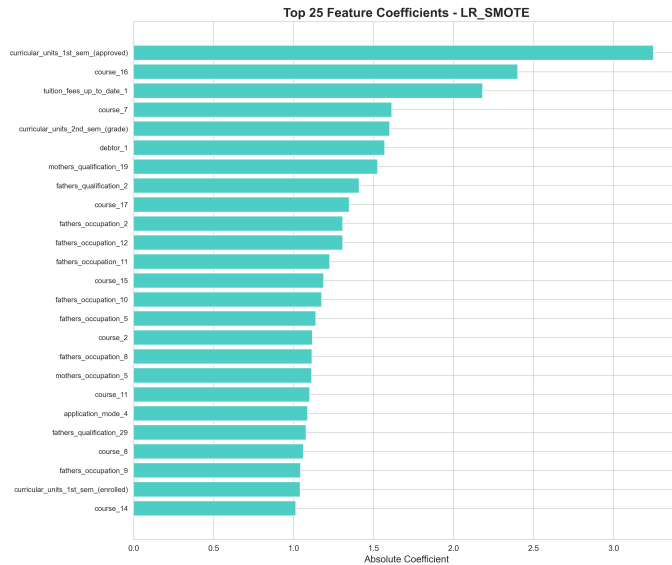


Fig. 4. Top 25 feature coefficients from the Logistic Regression with SMOTE model ranked by absolute coefficient value.

performance is three times more influential than other factors and providing specific, concrete advice on intervention timing and resource management. However, several limitations are to be acknowledged. Despite the dataset being of a good size, consisting of 4,424 students records, it is taken from a single Portuguese institution, which might put on a limit on how far these results can be generalized onto other educational contexts. The study also did not consider modeling techniques like recurrent neural networks or time-series methods, which could track the evolving nature of dropout risk over the academic year. Future studies should look into the optimal frequency and timing for prediction updates, and examine whether intervention effectiveness depends on specific risk profiles that are identified through feature importance analysis. Moreover, despite the study achieving strong performance metrics, the 10% of students who are at-risk remain unidentified, forming an important population that requires further investigation. Knowing why these students elude detection can lead to improved models.

A. Future Directions

Future studies should explore different directions to build upon these findings. First, temporal modeling approaches, such as recurrent neural networks or time-series analysis, could help with tracking how dropout risk changes over the course of an academic year. This could, in turn, help detect possible intervention windows even after the first semester. Second, conducting validation studies across institutions, educational settings, countries, and learning modalities would help confirm the generalizability of the results and identify institution-specific versus global dropout predictors. Third, making comprehensive cost-benefit models that quantify the economic trade-offs between false positives and false negatives would let institutions fine-tune decision thresholds in line

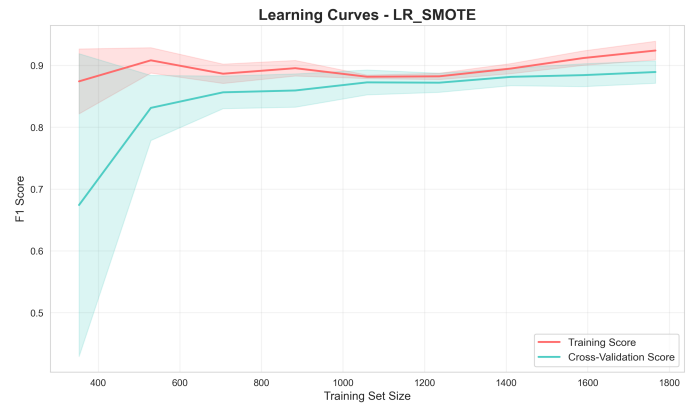


Fig. 5. Fig. 4. Learning curves for Logistic Regression with SMOTE showing F1 score as a function of training set size.

with their available intervention resources. Fourth, conducting qualitative analyses of the 10% of students who are at-risk and remain undetected could help identify edge cases that require additional or alternative approaches. Finally, longitudinal studies evaluating the effectiveness of interventions tailored to specific risk profiles, as shown by feature importance analysis, would help in bridging the gap between prediction and practice, resulting in tangible improvements in student retention outcomes.

V. CONCLUSION

This study demonstrates that interpretable machine learning models can achieve high predictive accuracy for student dropout prediction in online learning platforms while keeping interpretability intact. The Logistic Regression model, when trained with SMOTE oversampling, achieved an F1 score of 87.6% and identified 90% of at-risk students, retaining full interpretability through coefficient analysis. A systematic comparison of class weighting and SMOTE strategies delivers empirical evidence that SMOTE regularly exceeds class weighting across multiple model types, thereby tackling an important gap in the literature. Feature importance analysis shows that first-semester approved credits exert three times more influence than other predictors, pointing to the critical role of early academic performance monitoring. The results support the implementation of data-driven, early intervention strategies by enabling institutions to identify high-risk students early in their academic journey and to prioritize support resources for students with low first-semester credit approval, specific course enrollment patterns, and outstanding tuition payments. By combining high predictive performance accompanied by practical insights, this research offers an implementable solution for reducing dropout rates and enhancing student success in online learning environments.

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